Variational Free Energy (VFE) Telemetry:

Mathematical Formalization with Boltzmann Structural Prior

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Abstract

This document provides a rigorous mathematical formalization of the Variational Free Energy (VFE) telemetry system implemented in the MEGA framework. We establish formal definitions, prove key properties, and demonstrate how the VFE metric emerges naturally from MEGA's probabilistic dynamics as specified in the PFOL (Probabilistic First-Order Logic) theory. The complexity term is decomposed into two components: *selection complexity* measuring deviation from MEGA's natural equilibrium distribution, and *structural complexity* interpreted as the negative log-likelihood under a Boltzmann prior over hyperbolic space. This yields a principled three-term VFE decomposition that can be computed in natural units (nats) as well as normalized form.

Contents

1 Notation and Preliminaries **3**
1.1 State Space . . . 3
1.2 Organism Representation . . . 3
1.3 Delimiter Handling . . . 3
1.4 Live Meta Genes . . . 3

2 VFE Objective Function **3**
2.1 Core Definition . . . 3
2.2 Interpretative Framework . . . 4

3 Surprisal Term: Formalization **4**
3.1 Bigram Language Model . . . 4
3.2 Parent Selection . . . 4
3.3 Model Estimation . . . 5
3.4 Raw Surprisal Computation . . . 5
3.5 Normalized Surprisal . . . 5
3.6 Interpretation . . . 6

4 Complexity Term: Decomposition into Selection and Structural Components **6**
4.1 Natural Selection Distribution . . . 6
4.2 Empirical Distribution . . . 7
4.3 Selection Distribution on Active Vocabulary . . . 7
4.4 Selection Complexity: KL Divergence . . . 7
4.5 Boltzmann Structural Prior . . . 7
4.6 Recursive Computation of Meta Gene Statistics . . . 8
4.7 Total Complexity . . . 9
4.8 Normalized Complexity . . . 9
4.9 Interpretation of Complexity Components . . . 9

5 True VFE in Natural Units **10**
5.1 Three-Term Decomposition . . . 10
5.2 Properties of True VFE . . . 10

6 Connection to PFOL Axioms **10**
6.1 Surprisal and Population Dynamics . . . 10
6.2 Selection Complexity and Gene Selection (P3.1) . . . 11
6.3 Structural Complexity and Hyperbolic Geometry (A10.3) . . . 11
6.4 Meta Gene State Management (A3) . . . 11

7 Implementation Correspondence **11**
7.1 Code-to-Math Mapping . . . 11
7.2 Key Implementation Details . . . 11
7.2.1 Semantic Filtering . . . 11
7.2.2 Complexity Computation . . . 12
7.2.3 True VFE Computation . . . 12

8 Theoretical Properties **13**
8.1 Boundedness of Normalized VFE . . . 13
8.2 Equilibrium Characterization . . . 13
8.3 Non-Monotonicity . . . 13
8.4 Independence from Fitness . . . 14

9 Experimental Interpretation **14**
9.1 Diagnostic Use Cases . . . 14
9.1.1 Detecting Premature Convergence . . . 14
9.1.2 Measuring Exploration-Exploitation Balance . . . 14
9.1.3 Meta Gene Utility Assessment . . . 14
9.2 Telemetry Output . . . 15
9.3 Visualization . . . 15

10 Summary and Conclusions **15**
10.1 Key Contributions . . . 15
10.2 Theoretical Significance . . . 16
10.3 Relationship to Variational Inference . . . 16
10.4 Future Directions . . . 16
10.5 Final Remarks . . . 17

1 Notation and Preliminaries

1.1 State Space

Definition 1.1 (System State). *Let $\mathcal{S}(t) = (\mathcal{P}_t, \mathcal{E}(t), \mathcal{M}(t))$ denote the complete MEGA system state at generation $t \in \mathbb{N}$, where:*

- $\mathcal{P}_t = \{x_1, \ldots, x_N\}$*: Population of organisms where* $N = \pi_1$ *(population size)*
- $\mathcal{E}(t)$*: Encoding manager state, including hash encodings map and meta gene stack*
- $\mathcal{M}(t) = \mathcal{A}(t) \cup \mathcal{T}(t) \cup \mathcal{P}(t)$*: Meta gene state partition into Active, deletion baskeT, and reuse Pool sets*

1.2 Organism Representation

Definition 1.2 (Organism Encodings). *Each organism $x \in \mathcal{P}_t$ exists in two representations:*

1. ***Encoded form****:* $x = [g_1, g_2, \ldots, g_n]$ *where* $g_i \in \mathbb{E}$ *(may include meta gene references)*
2. ***Decoded form****:* $Decode(x)$ *= fully expanded sequence with all meta genes recursively replaced by their constituent sequences*

From PFOL axioms A2 and A5.3, we have the fundamental invariance:

Proposition 1.1 (Functional Invariance). *For all organisms $x \in \mathcal{X}$ and fitness function F:*

$$F(x) = F(Decode(x)) \tag{1}$$

1.3 Delimiter Handling

Definition 1.3 (Delimiter Stripping). *Define the delimiter stripping function $Strip : \mathcal{X} \to \mathcal{X}'$ as:*

$$Strip(x) = [g \in x \mid g \notin \{\varepsilon_{start}, \varepsilon_{end}\}] \tag{2}$$

where $\varepsilon_{start} = H(1)$ and $\varepsilon_{end} = H(2)$ are reserved delimiter encodings (PFOL A2.2).

This function removes structural delimiters while preserving semantic content, allowing focus on the actual gene sequence.

1.4 Live Meta Genes

Definition 1.4 (Live Meta Gene Set). *At generation t, define the set of live meta genes as:*

$$\mathcal{L}(t) = \{m \in \mathcal{A}(t) \mid \xi(m) \neq \emptyset \wedge \xi(m) \neq [m]\} \tag{3}$$

where $\xi(m)$ denotes the constituent sequence of meta gene m (excluding delimiters).

2 VFE Objective Function

2.1 Core Definition

Definition 2.1 (VFE Objective — Normalized Form). *The normalized Variational Free Energy at generation t is defined as:*

$$\mathcal{F}_{VFE}^{norm}(t) = \alpha \cdot Surprisal_{norm}(t) + \beta \cdot Complexity_{norm}(t) \tag{4}$$

where $\alpha, \beta \in [0, 1]$ are weights satisfying $\alpha + \beta = 1$ (typically both set to 0.5).

Definition 2.2 (True VFE — Natural Units)**.** *The true Variational Free Energy in nats is:*

$$\mathcal{F}_{VFE}^{nats}(t) = \underbrace{D_{KL}(q_{emp} \| q_{sel})}_{\textit{Selection Complexity}} + \underbrace{\mathbb{E}_{q_{emp}}[-\ln P_{struct}]}_{\textit{Structural Complexity}} + \underbrace{\mathbb{E}_{q_{emp}}[-\ln P_{gen}]}_{\textit{Accuracy (Surprisal)}} \tag{5}$$

This is an unbounded metric measured in nats that provides direct comparability across experiments.

Design Philosophy

The VFE metric is explicitly designed as a **passive telemetry signal**, not a fitness objective. It measures the natural dynamics of MEGA without influencing selection. This distinction is critical: VFE observes the system's behavior rather than directing it. The three-term decomposition corresponds to:

1. **Selection Complexity**: Deviation from MEGA's intrinsic selection prior (PFOL P3.1)
2. **Structural Complexity**: Cost of encoding complexity under a Boltzmann prior
3. **Accuracy**: Predictability of population transitions (surprisal)

2.2 Interpretative Framework

The three terms balance competing pressures:

- **Surprisal (Accuracy)**: Measures *predictability* of the population under the previous generation's generative model. High surprisal indicates rapid evolutionary change, exploration of novel combinations, or deviation from established patterns.
- **Selection Complexity**: Measures *deviation from MEGA's natural equilibrium* distribution as defined by its own selection operators (PFOL P3.1). High KL divergence indicates fitness pressure dominating neutral dynamics.
- **Structural Complexity**: Measures the *geometric cost* of the symbols used, interpreted as negative log-likelihood under a Boltzmann structural prior. High structural complexity indicates reliance on deeply nested, long meta genes.

3 Surprisal Term: Formalization

3.1 Bigram Language Model

Definition 3.1 (Conditional Probability Model)**.** *At generation t, define the bigram conditional probability model:*

$$P_t(b \mid a) = \Pr[\textit{next token is } b \mid \textit{current token is } a] \tag{6}$$

This model is estimated from the *selected parents* at generation t using maximum likelihood with Laplace smoothing.

3.2 Parent Selection

Definition 3.2 (Parent Set)**.** *The parent set $\mathcal{S}(\mathcal{P}_t)$ is deterministically selected via PFOL axiom P3 as the top π_2 organisms by fitness:*

$$\mathcal{S}(\mathcal{P}_t) = \{x \in \mathcal{P}_t \mid \textit{Rank}(x, t) \leq \pi_2\} \tag{7}$$

where $\textit{Rank}(x, t)$ denotes the fitness-based rank (1 = highest fitness).

These parents represent the "successful" patterns that the system propagates forward, forming the basis for the predictive model.

3.3 Model Estimation

Definition 3.3 (Bigram MLE with Smoothing)**.** *For alphabet Σ_t (all observed tokens at generation t), estimate:*

$$P_t(b \mid a) = \frac{Count_t(a, b) + \alpha_s}{Count_t(a, \cdot) + \alpha_s |\Sigma_t|} \tag{8}$$

where:

- $Count_t(a, b)$*: Number of bigram occurrences* $(a \to b)$ *in* $\{Strip(x) \mid x \in \mathcal{S}(\mathcal{P}_t)\}$
- $Count_t(a, \cdot) = \sum_{b \in \Sigma_t} Count_t(a, b)$
- $\alpha_s = 0.5$*: Laplace smoothing parameter*

3.4 Raw Surprisal Computation

Definition 3.4 (Negative Log-Likelihood per Transition)**.** *For organism x with $n = |Strip(x)|$ tokens, define:*

$$NLL_t(x) = -\sum_{i=2}^{n} \log P_t(Strip(x)[i] \mid Strip(x)[i-1]) \tag{9}$$

Definition 3.5 (Raw Surprisal)**.** *The raw surprisal at generation $t+1$ under model P_t is:*

$$Surprisal_{raw}(t+1) = \frac{1}{|\mathcal{P}_{t+1}|} \sum_{x \in \mathcal{P}_{t+1}} \frac{NLL_t(x)}{|Strip(x)| - 1} \tag{10}$$

Key observations:

1. Surprisal is computed on **delimiter-stripped** sequences to focus on semantic content
2. We normalize by organism length (number of transitions) to make surprisal comparable across different-length individuals
3. At $t = 0$, P_{-1} is undefined, so by convention $\text{Surprisal}_{\text{norm}}(0) = 1.0$

3.5 Normalized Surprisal

Definition 3.6 (Normalized Surprisal)**.** *To ensure comparability across generations with varying alphabet sizes:*

$$MaxEnt_t = \log |\Sigma_t| \tag{11}$$

$$Surprisal_{norm}(t) = \min\left(\frac{Surprisal_{raw}(t)}{MaxEnt_{t-1}}, 1.0\right) \tag{12}$$

This normalization bounds surprisal in $[0, 1]$:

- 0: Perfect predictability (population exactly follows parent model)
- 1: Maximum entropy (uniform random generation, no pattern preservation)

3.6 Interpretation

Remark 3.1 (Surprisal Interpretation)**.** *Surprisal measures how well the current population conforms to patterns learned from successful organisms in the previous generation:*

High Surprisal ($\to 1$) *indicates:*

- *Rapid evolutionary change (population structure shifting)*
- *Exploration of novel gene combinations*
- *Deviation from established successful patterns*

Low Surprisal ($\to 0$) *indicates:*

- *Convergence to stable patterns*
- *Exploitation of known successful structures*
- *Predictable population dynamics*

4 Complexity Term: Decomposition into Selection and Structural Components

Critical Principle

The complexity term decomposes into two fundamentally distinct components:

1. **Selection Complexity**: $D_{\mathrm{KL}}(q_{\mathrm{emp}} \| q_{\mathrm{sel}})$ — deviation from MEGA's natural equilibrium
2. **Structural Complexity**: $\mathbb{E}_{q_{\mathrm{emp}}}[d_H(\cdot)]$ — negative log-likelihood under Boltzmann prior

Neither component imposes external constraints; both measure intrinsic properties of MEGA's dynamics.

4.1 Natural Selection Distribution

From PFOL axiom P3.1, gene selection in MEGA follows:

Definition 4.1 (MEGA Gene Selection Distribution)**.** *Let $\mathcal{G}$ denote the base gene set and $\mathcal{M} = [m_0, \ldots, m_{|\mathcal{M}|-1}]$ the meta gene stack (LRU-ordered). The selection probability is:*

$$P_{\Pi}(G \mid \psi) = \begin{cases} \frac{\pi_{12}}{|\mathcal{G}|} & \text{if } G \in \mathcal{G} \text{ (base gene)} \\ \frac{(1-\pi_{12}) \cdot \pi_{13}^{|\mathcal{M}|-j-1}}{\sum_{k=0}^{|\mathcal{M}|-1} \pi_{13}^{|\mathcal{M}|-k-1}} & \text{if } G = m_j \in \mathcal{M} \text{ (meta gene)} \end{cases} \tag{13}$$

where:

- π_{12}*: Base gene probability mass (PFOL parameter* `base_gene_prob`*)*
- π_{13}*: Meta gene exponential decay factor (PFOL parameter* `metagene_prob`*)*
- j*: Position of meta gene in LRU stack (0 = oldest, $|\mathcal{M}| - 1$ = most recently used)*

4.2 Empirical Distribution

Definition 4.2 (Empirical Gene Distribution)**.** *The actual symbol distribution from population $\mathcal{P}_t$ is:*

$$q_{emp}(g) = \frac{\sum_{x \in \mathcal{P}_t} Count(g \text{ in } Strip(x))}{\sum_{x \in \mathcal{P}_t} |Strip(x)|} \tag{14}$$

This gives the frequency of each gene symbol in the delimiter-stripped population.

4.3 Selection Distribution on Active Vocabulary

Let $\Sigma = \text{supp}(q_{\text{emp}})$ be the *active vocabulary* (genes actually appearing in the population).

Definition 4.3 (Restricted Selection Distribution)**.** *For the active vocabulary Σ, define:*

For base genes $g \in \Sigma \cap \mathcal{G}$*:*

$$q_{sel}(g) = \frac{\pi_{12}}{|\Sigma \cap \mathcal{G}|} \tag{15}$$

For meta genes $m_j \in \Sigma \cap \mathcal{M}$*:*

$$q_{sel}(m_j) = \frac{(1 - \pi_{12}) \cdot \pi_{13}^{|\mathcal{M}|-j-1}}{Z_\Sigma} \tag{16}$$

where:

$$Z_\Sigma = \sum_{m_k \in \Sigma \cap \mathcal{M}} \pi_{13}^{|\mathcal{M}|-k-1} \tag{17}$$

Then normalize over Σ:

$$q_{sel}(g) \leftarrow \frac{q_{sel}(g)}{\sum_{g' \in \Sigma} q_{sel}(g')} \tag{18}$$

4.4 Selection Complexity: KL Divergence

Definition 4.4 (Selection Complexity)**.** *The selection complexity measures deviation from MEGA's natural equilibrium:*

$$\mathcal{C}_{sel} = D_{KL}(q_{emp} \| q_{sel}) = \sum_{g \in \Sigma} q_{emp}(g) \log \left(\frac{q_{emp}(g)}{q_{sel}(g)} \right) \tag{19}$$

Properties:

- $\mathcal{C}_{\text{sel}} = 0$: Perfect equilibrium ($q_{\text{emp}} = q_{\text{sel}}$)
- $\mathcal{C}_{\text{sel}} > 0$: Fitness pressure warping the natural distribution
- Units: nats (natural logarithm)

4.5 Boltzmann Structural Prior

Key Innovation: Boltzmann Interpretation

The hyperbolic cost function from PFOL A10.3 is interpreted as defining a **Boltzmann structural prior**:

$$P_{\text{struct}}(g) \propto \exp(-d_H(g)) \tag{20}$$

where $d_H(g)$ is the hyperbolic distance of symbol g from the origin. The structural complexity is then the *negative log-likelihood* under this prior:

$$-\log P_{\text{struct}}(g) = d_H(g) + \text{const} \tag{21}$$

Definition 4.5 (Hyperbolic Distance Function)**.** *For meta gene $m \in \mathcal{L}(t)$, define the hyperbolic distance from origin:*

$$d_H(m) = \mathit{arccosh}(1 + \mathit{depth}(m)) \cdot \mathit{flatlen}(m) \tag{22}$$

where:

- *$\mathit{depth}(m)$: Maximum nesting depth (number of recursive meta gene levels)*
- *$\mathit{flatlen}(m)$: Terminal-expanded length (all meta genes recursively replaced)*
- *$\mathit{arccosh}(1 + d)$: Hyperbolic distance scaling from PFOL A10.3*

For base genes: $d_H(g) = 0$ (no internal structure, located at origin).

Definition 4.6 (Structural Complexity)**.** *The structural complexity is the expected hyperbolic distance under the empirical distribution:*

$$\mathcal{C}_{\mathit{struct}} = \mathbb{E}_{q_{\mathit{emp}}}[d_H] = \sum_{g \in \Sigma} q_{\mathit{emp}}(g) \cdot d_H(g) \tag{23}$$

Under the Boltzmann interpretation:

$$\mathcal{C}_{\text{struct}} = \mathbb{E}_{q_{\text{emp}}}[-\log P_{\text{struct}}] + \text{const} \tag{24}$$

This is precisely the expected negative log-likelihood of the empirical distribution under the structural prior.

4.6 Recursive Computation of Meta Gene Statistics

Algorithm 1 Compute Meta Gene Flat Length and Depth

```
function FlattenMetaStats(m, ℰ, cache)
    if m ∈ cache then
        return cache[m]
    end if
    ξ ← ℰ.encodings[m]                                  ▷ Constituent sequence
    if ξ = ∅ or ξ = [m] then
        return (1.0, 0.0)                               ▷ Terminal: length 1, depth 0
    end if
    total_len ← 0.0
    max_child_depth ← 0.0
    for each c ∈ ξ do
        if c is meta gene then
            (ℓ, d) ← FlattenMetaStats(c, ℰ, cache)
            total_len ← total_len + ℓ
            max_child_depth ← max(max_child_depth, d)
        else
            total_len ← total_len + 1
        end if
    end for
    depth ← 1 + max_child_depth
    cache[m] ← (total_len, depth)
    return (total_len, depth)
end function
```

4.7 Total Complexity

Definition 4.7 (Raw Complexity)**.** *The raw complexity is the sum of selection and structural components:*

$$Complexity_{raw} = \mathcal{C}_{sel} + \mathcal{C}_{struct} = D_{KL}(q_{emp} \| q_{sel}) + \mathbb{E}_{q_{emp}}[d_H] \tag{25}$$

4.8 Normalized Complexity

Definition 4.8 (Normalized Complexity)**.** *To map complexity to* $[0, 1]$*, define:*

$$Scale = \log |\Sigma| + 1 \tag{26}$$

$$Complexity_{norm} = 1 - \exp\left(-\frac{Complexity_{raw}}{Scale}\right) \tag{27}$$

This produces a smooth, bounded measure:

- 0: Perfect alignment with MEGA's natural dynamics and minimal structure
- 1: Extreme deviation from equilibrium and/or high structural complexity

4.9 Interpretation of Complexity Components

Remark 4.1 (Selection Complexity Interpretation)**.** ***High*** $\mathcal{C}_{sel}$ *indicates:*

- *Fitness pressure strongly favoring certain genes beyond their LRU position*
- *Population structure warped away from neutral equilibrium*
- *Strong selection overriding MEGA's intrinsic balance*

Low $\mathcal{C}_{sel}$ *indicates:*

- *System near its neutral equilibrium*
- *Fitness and natural selection pressures in balance*
- *Gene usage respects LRU decay structure*

Remark 4.2 (Structural Complexity Interpretation)**.** ***High*** $\mathcal{C}_{struct}$ *indicates:*

- *Heavy reliance on deeply nested meta genes*
- *Large terminal-expanded sequences being compressed*
- *Complex hierarchical structure in the population*

Low $\mathcal{C}_{struct}$ *indicates:*

- *Predominantly base gene usage*
- *Shallow meta gene hierarchies*
- *Simple, direct encodings*

5 True VFE in Natural Units

5.1 Three-Term Decomposition

Definition 5.1 (True VFE (nats)). *The true Variational Free Energy in natural units combines all three terms:*

$$\mathcal{F}_{VFE}^{nats}(t) = \mathcal{C}_{sel}(t) + \mathcal{C}_{struct}(t) + Surprisal_{raw}(t) \tag{28}$$

Equivalently:

$$\mathcal{F}_{VFE}^{nats}(t) = D_{KL}(q_{emp} \| q_{sel}) + \mathbb{E}_{q_{emp}}[d_H] + \mathbb{E}[-\log P_{gen}] \tag{29}$$

VFE as Evidence Lower Bound

The true VFE has a natural interpretation as a variational bound:

$$\mathcal{F}_{\text{VFE}}^{\text{nats}} = \underbrace{D_{\text{KL}}(q \| p_{\text{sel}})}_{\text{Selection Prior Divergence}} + \underbrace{\mathbb{E}_q[-\log P_{\text{struct}}]}_{\text{Structural Prior NLL}} + \underbrace{\mathbb{E}_q[-\log P_{\text{gen}}]}_{\text{Generative Model NLL}} \tag{30}$$

This mirrors the structure of variational inference where:

- $q = q_{\text{emp}}$: The empirical (variational) distribution
- p_{sel}: MEGA's intrinsic selection prior
- P_{struct}: Boltzmann structural prior
- P_{gen}: Generative transition model (bigram)

5.2 Properties of True VFE

Theorem 5.1 (Non-negativity). *For all generations t:*

$$\mathcal{F}_{VFE}^{nats}(t) \geq 0 \tag{31}$$

Proof. Each component is non-negative:

- $D_{\text{KL}}(q \| p) \geq 0$ by Gibbs' inequality
- $\mathbb{E}[d_H] \geq 0$ since $d_H(g) \geq 0$ for all g
- $\text{Surprisal}_{\text{raw}} \geq 0$ since $-\log p \geq 0$ for $p \in (0, 1]$

□

Remark 5.1 (Unboundedness). *Unlike the normalized VFE, $\mathcal{F}_{VFE}^{nats}$ is unbounded above. This allows direct comparison of VFE magnitudes across different experiments and parameter settings.*

6 Connection to PFOL Axioms

6.1 Surprisal and Population Dynamics

Proposition 6.1 (Surprisal as Generative Likelihood). *The surprisal term directly implements the generative likelihood from PFOL A9.4. Specifically, under the bigram approximation:*

$$\log P_{\Pi}(o_i \mid parents, \Pi) \approx -NLL_t(o_i) \tag{32}$$

Thus, surprisal is the average negative log-likelihood of offspring under the bigram model of the parent distribution.

6.2 Selection Complexity and Gene Selection (P3.1)

Proposition 6.2 (Selection Distribution Correspondence)**.** *The selection distribution q_{sel} is constructed exactly from PFOL P3.1:*

$$q_{sel} \propto P_{\Pi}(G \mid \psi) \tag{33}$$

restricted to the active vocabulary Σ.

The parameters π_{12} and π_{13} are extracted directly from the MEGA configuration, ensuring mathematical consistency with the PFOL specification.

6.3 Structural Complexity and Hyperbolic Geometry (A10.3)

Proposition 6.3 (Structural Cost as Hyperbolic Distance)**.** *The structural complexity function implements PFOL A10.3 hyperbolic distance:*

$$d_H(m) = d_{\mathbb{H}}(\phi(m), \mathit{origin}) \cdot \mathit{flatlen}(m) \tag{34}$$

where:

- $d_{\mathbb{H}}(x, y) = \mathit{arccosh}(-\langle x, y\rangle_M)$ *is the hyperbolic distance*
- $\phi : \mathcal{X} \to \mathbb{H}^n$ *is the embedding into hyperbolic space*
- *For depth d:* $d_{\mathbb{H}} \approx \mathit{arccosh}(1 + d)$

The Boltzmann interpretation $P_{\text{struct}}(m) \propto \exp(-d_H(m))$ naturally penalizes symbols far from the origin in hyperbolic space, where complexity grows exponentially with depth.

6.4 Meta Gene State Management (A3)

Proposition 6.4 (LRU Consistency)**.** *The LRU ordering in $\mathcal{M}$ determines q_{sel} via the index j in Definition 4.1. Usage updates (PFOL A3.3) directly influence the selection distribution through position changes in the stack.*

This creates a feedback loop:

1. Meta genes used in high-fitness organisms get higher LRU positions
2. Higher LRU positions increase q_{sel} weight
3. Deviation from this natural weighting increases selection complexity

7 Implementation Correspondence

7.1 Code-to-Math Mapping

7.2 Key Implementation Details

7.2.1 Semantic Filtering

The implementation enforces PFOL's semantic token space assumption:

```
function SemanticFilter(pop_stripped, ga)
    G_ids ← {H(g) | g ∈ ga.genes}                ▷ Base gene hash IDs
    M_ids ← meta_gene_stack                      ▷ Live meta gene IDs
    pop_sem ← []
    for each seq in pop_stripped do
```

Code Variable	Mathematical Symbol
`pop_sem`	$\{\text{Strip}(x) \mid x \in \mathcal{P}_t\} \cap (\mathcal{G} \cup \mathcal{M})$
`self._prev_Pcond`	P_t (bigram model)
`q_emp`	q_{emp}
`q_sel`	q_{sel}
`structural_cost_map`	$\{d_H(m) \mid m \in \Sigma \cap \mathcal{M}\}$
`kl_emp_sel`	$\mathcal{C}_{\text{sel}} = D_{\text{KL}}(q_{\text{emp}} \| q_{\text{sel}})$
`structural_complexity`	$\mathcal{C}_{\text{struct}} = \mathbb{E}_{q_{\text{emp}}}[d_H]$
`true_vfe_nats`	$\mathcal{F}_{\text{VFE}}^{\text{nats}}$
`pi12` / `base_gene_prob`	π_{12}
`pi13` / `metagene_prob`	π_{13}
`live_meta_ids`	$\mathcal{L}(t)$
`meta_gene_stack`	$\mathcal{M}$ (LRU-ordered)

Table 1: Code-to-mathematics correspondence

```
        pop_sem.append([t ∈ seq | t ∈ G_ids ∪ M_ids])
    end for
    return pop_sem, G_ids, M_ids
end function
```

7.2.2 Complexity Computation

```
function ComputeComplexity(ga, pop_sem, meta_set)
    Compute q_emp from pop_sem
    Extract π_12, π_13 from ga config
    Construct q_sel per Definition 4.1
    Normalize q_sel over Σ = supp(q_emp)

                                        ▷ Structural complexity via Boltzmann prior
    meta_stats ← FlattenMetaStats(meta_set)
    for each g ∈ Σ do
        if g ∈ G then
            structural_cost[g] ← 0
        else if g ∈ meta_stats then
            (flatlen, depth) ← meta_stats[g]
            structural_cost[g] ← arccosh(1 + depth) × flatlen
        end if
    end for

    C_sel ← D_KL(q_emp || q_sel)
    C_struct ← Σ_{g∈Σ} q_emp(g) · structural_cost[g]
    complexity_raw ← C_sel + C_struct

    scale ← log |Σ| + 1
    complexity_norm ← 1 − exp(−complexity_raw/scale)
    return C_sel, C_struct, complexity_raw, complexity_norm
end function
```

7.2.3 True VFE Computation

```
function ComputeTrueVFE(C_sel, C_struct, surprisal_raw)
```

```
    true_vfe_nats ← C_sel + C_struct + surprisal_raw
    return true_vfe_nats
end function
```

8 Theoretical Properties

8.1 Boundedness of Normalized VFE

Theorem 8.1 (Normalized VFE Boundedness)**.** *For all generations t:*

$$0 \leq \mathcal{F}_{VFE}^{norm}(t) \leq 1 \tag{35}$$

Proof. Both normalized components are bounded:

- $0 \leq \text{Surprisal}_{\text{norm}}(t) \leq 1$ by definition (min operation)
- $0 \leq \text{Complexity}_{\text{norm}}(t) \leq 1$ by exponential mapping

Since $\alpha, \beta \in [0, 1]$ and $\alpha + \beta = 1$:

$$\mathcal{F}_{\text{VFE}}^{\text{norm}}(t) = \alpha \cdot [\text{bounded in } [0, 1]] + \beta \cdot [\text{bounded in } [0, 1]] \in [0, 1] \tag{36}$$

□

8.2 Equilibrium Characterization

Definition 8.1 (VFE Equilibrium)**.** *A system is in* VFE equilibrium *at generation t if:*

$$\mathcal{F}_{VFE}^{nats}(t) \to 0 \tag{37}$$

Proposition 8.2 (Equilibrium Conditions)**.** *VFE equilibrium implies all three conditions simultaneously:*

1. $\mathcal{C}_{sel} \to 0$*: $q_{emp} = q_{sel}$ (population matches MEGA's intrinsic balance)*
2. $\mathcal{C}_{struct} \to 0$*: Population uses only base genes (no meta gene complexity)*
3. $Surprisal_{raw} \to 0$*: Population perfectly predicted by parent model*

Remark 8.1 (Practical Implications)**.** *True equilibrium is rarely achieved in practice because:*

- *Fitness pressure typically drives q_{emp} away from q_{sel}*
- *Useful compression via meta genes increases $\mathcal{C}_{struct}$*
- *Mutation and crossover introduce surprisal*

A "healthy" evolutionary trajectory balances these pressures rather than minimizing VFE.

8.3 Non-Monotonicity

Remark 8.2 (VFE Trajectory Properties)**.** *Unlike fitness (which evolutionary algorithms seek to maximize), VFE is not monotonic:*

- ***Exploration phases**: VFE increases (high surprisal, deviation from equilibrium)*
- ***Exploitation phases**: VFE decreases (convergence, return to equilibrium)*
- ***Optimal trajectories**: Exhibit oscillation between exploration and exploitation*

8.4 Independence from Fitness

Theorem 8.3 (Fitness Independence)**.** *The VFE metric is computed independently of the fitness function F:*

$$\mathcal{F}_{VFE}(t) = \mathcal{F}_{VFE}(\mathcal{P}_t, \mathcal{E}(t), P_{t-1}, \Pi) \tag{38}$$

where fitness F does not appear explicitly.

Proof. By construction:

- Surprisal depends only on: population $\mathcal{P}_t$, previous model P_{t-1}
- Selection complexity depends on: population $\mathcal{P}_t$, parameters Π
- Structural complexity depends on: population $\mathcal{P}_t$, encoding state $\mathcal{E}(t)$
- No term references fitness values directly

Fitness influences VFE only indirectly through selection (which organisms become parents), not through direct computation. □

9 Experimental Interpretation

9.1 Diagnostic Use Cases

9.1.1 Detecting Premature Convergence

- **Symptom**: $\text{Surprisal}_{\text{norm}} \to 0$ while fitness plateaus below optimum
- **Interpretation**: Population has converged to suboptimal local structure with high predictability
- **Recommendation**: Increase mutation rates or introduce diversity mechanisms

9.1.2 Measuring Exploration-Exploitation Balance

- **High VFE**: System exploring (high surprisal) or deviating from equilibrium (high complexity)
- **Low VFE**: System exploiting known patterns in balance with natural dynamics
- **Optimal**: VFE oscillates, indicating healthy balance between phases

9.1.3 Meta Gene Utility Assessment

- Track: $\mathcal{C}_{\text{sel}}$ and $\mathcal{C}_{\text{struct}}$ separately
- **High $\mathcal{C}_{\text{sel}}$, Low $\mathcal{C}_{\text{struct}}$**: Fitness favoring specific base genes
- **Low $\mathcal{C}_{\text{sel}}$, High $\mathcal{C}_{\text{struct}}$**: Complex meta genes in natural balance
- **High both**: Strong selection for complex hierarchical structures

9.2 Telemetry Output

The implementation logs per generation:

- `true_vfe_nats`: $\mathcal{F}_{\text{VFE}}^{\text{nats}}$ (unbounded, in nats)
- `surprisal_norm`, `surprisal_raw`
- `complexity_norm`, `complexity_raw`
- `kl_emp_sel`: $\mathcal{C}_{\text{sel}}$ (selection complexity)
- `structural_complexity`: $\mathcal{C}_{\text{struct}}$ (structural complexity)
- `active_vocab_size`: $|\Sigma|$
- `vfe`: $\mathcal{F}_{\text{VFE}}^{\text{norm}}$ (normalized, bounded)
- `best_fitness`, `mean_fitness` (for correlation analysis)
- `meta_births`, `meta_deaths` (meta gene dynamics)
- `meta_live`, `meta_avg_len`, `meta_max_len` (structural metrics)

9.3 Visualization

The `VFETelemetry.plot()` method generates four subplots:

1. Surprisal over time (normalized)
2. Complexity over time (normalized)
3. Normalized VFE objective over time
4. Fitness comparison (best vs mean)

These allow visual inspection of:

- Correlation/anti-correlation between VFE and fitness
- Phase transitions (exploration $\leftrightarrow$ exploitation)
- System stability (variance in VFE)

10 Summary and Conclusions

10.1 Key Contributions

This formalization establishes:

1. **Rigorous three-term decomposition** of VFE into selection complexity, structural complexity, and accuracy (surprisal)
2. **Boltzmann structural prior interpretation**: The hyperbolic cost function is formally connected to a Boltzmann distribution over meta gene complexity, making structural complexity a proper negative log-likelihood
3. **True VFE in natural units**: An unbounded metric $\mathcal{F}_{\text{VFE}}^{\text{nats}}$ that allows direct comparison across experiments

4. **PFOL integration**: Direct correspondence between VFE components and PFOL axioms (P3.1 for selection, A10.3 for geometry, A9.4 for dynamics)

5. **Implementation verification**: Code-to-math mapping showing exact correspondence between Python implementation and formal specifications

10.2 Theoretical Significance

The VFE metric provides:

- **Passive observation window** into MEGA's dynamics without influencing evolution
- **Three complementary lenses**: Selection deviation, structural complexity, and predictability
- **Principled priors**: Uses MEGA's own equilibrium and hyperbolic geometry, not external biases
- **Diagnostic power**: Enables detection of convergence, phase transitions, meta gene utility

10.3 Relationship to Variational Inference

The true VFE has the structure of a variational free energy bound:

$$\mathcal{F} = D_{\mathrm{KL}}(q\|p) + \mathbb{E}_q[-\log p_{\mathrm{prior}}] + \mathbb{E}_q[-\log p_{\mathrm{likelihood}}] \tag{39}$$

In our context:

- $q = q_{\mathrm{emp}}$: Empirical distribution (variational approximation)
- $p = q_{\mathrm{sel}}$: MEGA's selection prior
- $p_{\mathrm{prior}} = P_{\mathrm{struct}}$: Boltzmann structural prior
- $p_{\mathrm{likelihood}} = P_{\mathrm{gen}}$: Generative transition model

This connects MEGA's evolutionary dynamics to the broader framework of variational inference and the Free Energy Principle.

10.4 Future Directions

Potential extensions:

1. **Higher-order generative models**: Extend beyond bigrams to capture longer-range dependencies
2. **Adaptive weighting**: Learn α, β from trajectory data
3. **Causal VFE**: Incorporate causal structure of meta gene dependencies
4. **Predictive tuning**: Use VFE trajectory to adaptively adjust Π parameters
5. **Cross-domain benchmarking**: Apply VFE to other evolutionary systems

10.5 Final Remarks

The VFE telemetry system provides a principled, mathematically grounded framework for understanding MEGA's evolutionary dynamics. By decomposing complexity into selection and structural components, and interpreting the latter through a Boltzmann prior, we obtain a metric with clear theoretical foundations and practical utility.

Core Insight: VFE is not a fitness function—it's a thermometer measuring the balance between MEGA's intrinsic dynamics and the pressures imposed by fitness-driven selection. The Boltzmann structural prior provides a principled way to quantify the "cost" of hierarchical compression, completing the connection between information theory, geometry, and evolutionary dynamics.